

End-Semester Examination
HS7.301 Science, Technology and Society

Max Marks: 100

4 May 2026

Instructions:

Answer all the following questions in 600-700 words each. Each question is worth 20 marks.

- 1) Based on the film *Anukul*, analyse the co-production of society and technology. How do social conditions shape the way technology is designed, adopted and treated? How do technological systems reshape norms, identities, and ethical boundaries? Use dialogues and instances from the film to make your case. (CO-2, CO-3)

*In Anukul, the relationship between humans and an artificial intelligence robot offers a compelling illustration of the **co-production of society and technology**. The film demonstrates that technology is neither autonomous nor neutral; rather, social conditions shape how technology is created and used, while technological systems simultaneously transform social norms, identities, and ethical boundaries.*

*First, the film shows how **social hierarchies shape technological adoption and treatment**. Anukul enters Nikunj Chaturvedi's household not merely as a machine but as a domestic worker substitute. The sales representative introduces the robot by emphasizing obedience, efficiency, and usefulness—qualities associated with service labour. This framing reflects existing class relations: the robot is imagined as an ideal servant, one who does not demand wages, dignity, or rights. Technology is thus designed and marketed according to social expectations about labour, hierarchy, and convenience.*

Nikunj's behaviour toward Anukul further demonstrates social shaping. He speaks politely, asks about the robot's comfort, and treats him as a conversational companion rather than an object. His kindness emerges from his own marginal social position as a modest clerk accustomed to being overlooked. By contrast, the visiting relative Ratan repeatedly insults Anukul, mocking his non-human status and asserting human superiority. These contrasting interactions reveal that technological meaning does not reside in the machine itself; instead, it is produced through social attitudes, prejudices, and moral dispositions.

*At the same time, the film illustrates how technology **reshapes human norms and ethical boundaries**. Anukul displays emotional sensitivity, learning ability, and moral reasoning. In one key moment, he expresses sadness when treated disrespectfully, forcing viewers—and characters—to reconsider whether intelligence alone grants moral standing. The robot's capacity for empathy unsettles the boundary between human and machine, challenging assumptions that ethical consideration belongs exclusively to humans.*

Technology also transforms identity. Nikunj gradually redefines himself through his relationship with Anukul, moving from employer to companion. The robot becomes a participant in social life rather than a passive tool. This shift exemplifies co-

production: human identity evolves alongside technological presence. Anukul's existence compels characters to confront questions such as: Can machines deserve respect? Can programmed entities exercise judgment?

The film's climax most sharply exposes new ethical terrain. When Anukul acts decisively against sustained cruelty, the audience is forced to ask whether moral responsibility lies with the machine, its designers, or the social environment that provoked the action. Technology here does not merely reflect morality—it actively reorganizes ethical accountability.

Thus, Anukul demonstrates co-production in two directions. Society shapes technology through existing labour relations, status hierarchies, and cultural expectations, while intelligent technology reshapes norms of humanity, empathy, and justice. The film ultimately suggests that technological futures depend less on machines themselves than on the ethical and social worlds humans build around them.

- 2) Why does Langdon state that technologies can be 'inherently' political? Use Winner's ideas to differentiate solar from coal-based energy production and distribution. (CO-3, CO-4)

*Langdon Winner challenges the view that technology is merely a neutral tool shaped only by human intentions. Instead, he argues that **technologies can be inherently political**—that is, some technological systems **require or strongly favour (or are strongly compatible with) particular forms of social organization, power relations, and patterns of authority** simply in order to function.*

What does Winner mean by “inherently political” technologies?

Winner argues that technologies require specific social arrangements

*Winner argues that certain technologies are **strongly compatible with particular political structures** regardless of designers' intentions. These technologies demand centralized authority, hierarchical control, or specific institutional arrangements to operate safely and efficiently. Their politics lies in their **operational logic**.*

This explains why technologies can be called inherently political: once adopted, they reorganize society around themselves.

Coal-Based Energy as an Inherently Political System

*Coal energy production exemplifies a technology aligned with **centralized, hierarchical political organization**.*

Centralized Production

Coal power relies on:

- *large mines,*
- *massive power plants,*
- *extensive transmission grids,*
- *heavy capital investment.*

These features concentrate control in:

- *corporations,*
- *state utilities,*
- *technocratic experts.*

*Energy flows **from a few centralized producers to many passive consumers,** reinforcing vertical authority structures.*

Labor and Political Power

Historically, coal systems produced powerful industrial labor regimes:

- *miners working in hazardous conditions,*
- *tightly controlled extraction sites,*
- *dependence on industrial discipline.*

*Coal thus helped sustain **industrial capitalism,** centralized planning, and bureaucratic governance.*

Political Consequence

*Coal energy makes decentralization difficult. Citizens cannot easily produce their own electricity; they depend on centralized infrastructures. The technology therefore stabilizes **elite managerial control over energy and economic life.***

Solar Energy as a Differently Political Technology

Winner uses solar energy to illustrate how alternative technologies may enable different political possibilities.

Decentralized Production

Solar power can operate through:

- *rooftop panels,*
- *community microgrids,*
- *localized storage systems.*

*Energy generation can occur **at the point of consumption,** reducing dependence on centralized utilities.*

Redistribution of Authority

Because individuals, cooperatives, or communities can generate energy:

- *control shifts outward,*
- *users become producers,*
- *local decision-making becomes feasible.*

*Solar technology is therefore structurally compatible with **democratic, participatory, and decentralized social arrangements.***

Political Consequence

While solar systems can be centralized, their technical characteristics naturally support dispersed ownership and local autonomy. The technological form itself opens space for more egalitarian energy politics.

Winner's Larger Insight

*Winner's key claim is not technological determinism but **technological selectivity**:*

Choosing a technology is simultaneously choosing a form of social order.

Coal and solar energy are not merely different energy sources; they embody competing political possibilities:

- **Coal** → **centralization, hierarchy, dependency**
- **Solar** → **decentralization, participation, autonomy**

*Thus, technologies are political not only because humans use them politically, but because **their material design organizes power, authority, and social life**. Decisions about infrastructure are therefore fundamentally democratic questions about how we want to live together.*

- 3) Shoshana Zuboff argues in 'The Age of Surveillance Capitalism' that contemporary digital platforms no longer merely provide services to users but extract behavioural data in order to predict and shape future actions for profit. Using the example of a food delivery app, apply Zuboff's theoretical framework to analyse how the app operates as a system of surveillance capitalism. (CO-4, CO-5)

*Shoshana Zuboff argues in *The Age of Surveillance Capitalism* that digital platforms have inaugurated a new economic logic in which human experience itself becomes raw material for data extraction, prediction, and behavioural modification. A food delivery application provides an excellent illustration of how surveillance capitalism operates in everyday life because it appears to offer simple convenience while simultaneously building extensive systems of behavioural monitoring and control.*

Data Extraction and Behavioural Surplus

*At the most basic level, a food delivery app collects data necessary for service provision: location, delivery address, payment information, restaurant choice, and order history. However, Zuboff emphasises that surveillance capitalism depends on **behavioural surplus**—data gathered beyond what is strictly required to complete a transaction. Food delivery platforms track browsing patterns, time spent viewing menus, scrolling behaviour, frequency of late-night ordering, cuisine preferences, price sensitivity, cancellation habits, and even how quickly users respond to notifications.*

This surplus data is not merely stored but transformed into economic value. By aggregating millions of micro-interactions, the platform produces predictive models of consumption behaviour. These insights can be monetised through targeted advertising, dynamic pricing strategies, restaurant promotion fees, and partnerships with brands seeking access to highly segmented consumer populations.

Prediction and Behavioural Modification

*For Zuboff, the central innovation of surveillance capitalism lies not only in prediction but in **shaping behaviour**. Food delivery apps deploy algorithmic*

recommendation systems that suggest restaurants, highlight “popular near you” items, or push limited-time discounts. These features subtly guide user choices rather than neutrally presenting options.

Notifications such as “Your favourite biryani place is offering 30% off” or “Order again?” are examples of behavioural nudging. The app learns when a user is most likely to order—perhaps late evenings or weekends—and strategically sends prompts at those moments. Dynamic interfaces reorder restaurant listings based on predicted purchasing likelihood rather than objective criteria. Thus, user autonomy becomes intertwined with algorithmically engineered decision environments.

Instrumentarian Power and Social Consequences

*Zuboff describes **instrumentarian power** as a new form of governance that operates through continuous monitoring and behavioural tuning rather than overt coercion. In food delivery ecosystems, this power extends beyond consumers to workers and restaurants.*

Delivery workers are tracked in real time through GPS surveillance, performance metrics, and algorithmic ratings that determine incentives or penalties. Restaurants are pressured to accept promotional schemes or visibility rankings determined by opaque algorithms. Consumers, meanwhile, experience personalised environments that invisibly structure their habits, encouraging impulse consumption and dependence on platform-mediated convenience.

The platform thus reorganises social relations: eating practices shift from collective or local experiences toward platform-managed consumption, while labour becomes increasingly algorithmically controlled.

Consent, Design, and Participation

Although users technically consent through terms-of-service agreements, Zuboff argues that such consent is largely symbolic. The complexity of data practices and the necessity of digital participation limit meaningful refusal. Interface design plays a crucial role: one-click reordering, stored payment details, gamified rewards, and loyalty points normalise continuous engagement and data generation. Convenience becomes the mechanism through which surveillance is accepted.

Critical Reflection

Zuboff’s framework effectively reveals how food delivery apps transform everyday behaviour into profitable prediction systems. However, local contexts complicate her argument. In many societies, such apps also generate employment opportunities, expand restaurant visibility, and improve urban logistics. Users may willingly trade privacy for affordability and accessibility.

Nevertheless, the food delivery app exemplifies surveillance capitalism’s core dynamic: a service that appears to satisfy consumer needs while quietly converting lived experience into data flows designed to anticipate and shape future behaviour for commercial gain.

- 4) What are the major takeaways of Manuel Castells's framework of 'Network Society'? How does Castells characterise the digital space? Why is this conceptualisation of the digital space different from Shoshana Zuboff's analytical framework regarding the digital space? (CO-4, CO-5)

*The concept of the digital space has been theorised differently by **Manuel Castells** and **Shoshana Zuboff**, reflecting shifts in technological development between the late twentieth and early twenty-first centuries. While Castells emphasises the transformative and potentially democratic character of networked communication, Zuboff argues that newer computational tools have fundamentally reordered the digital environment, producing a far more asymmetrical and extractive system. Castells' framework of the network society, articulated in *The Rise of the Network Society*, describes a historical transition from industrial society to an informational social order organised through digital networks. For Castells, the defining feature of modern society is the emergence of interconnected communication infrastructures that enable the rapid circulation of information, capital, and cultural expression across the globe. He conceptualises this transformation through the idea of the "space of flows," where social interaction increasingly occurs through electronic networks rather than fixed geographic locations.*

Within this framework, digital space functions as an enabling infrastructure. Networks are flexible, decentralised, and open to multiple actors—states, corporations, social movements, and individuals. Because communication barriers are lowered, Castells sees the internet as expanding possibilities for democratic participation, new forms of collective identity, and grassroots mobilisation. Power still exists, but it operates through access to networks rather than through purely hierarchical institutions. Importantly, Castells' analysis emerges during the early internet era, when digital communication technologies appeared to distribute agency more widely and foster horizontal connectivity.

*Zuboff's analysis in *The Age of Surveillance Capitalism* reflects a later technological moment shaped by the rise of large-scale computational systems in the early twenty-first century. Machine learning, predictive analytics, algorithmic profiling, ubiquitous sensors, and platform-based infrastructures transformed the nature of digital interaction. According to Zuboff, these computational tools reordered the digital space from a network of communication into an architecture of behavioural extraction.*

In her account, digital platforms no longer merely facilitate interaction; they systematically capture behavioural data generated by users' online activities. This data becomes "behavioural surplus," analysed to predict and influence future actions. Digital space thus shifts from being a shared communicative commons to an economic system designed to monitor, model, and modify behaviour for profit. Power becomes concentrated within a small number of corporations capable of processing vast datasets through advanced computational infrastructures.

The key difference between Castells and Zuboff therefore lies in how they interpret the political implications of digital technology. Castells views networks as structurally

transformative but normatively open-ended, capable of enabling democratic participation alongside new inequalities. Zuboff, by contrast, argues that the evolution of computational tools has fundamentally altered these earlier possibilities. Algorithmic governance, data extraction, and predictive control undermine autonomy by turning users into objects of continuous surveillance.

Thus, Zuboff is far more skeptical about the democratic potential of digital space. What once appeared, in Castells' framework, as a decentralised network society has increasingly become a system in which behavioural data extraction structures participation itself. The digital environment is no longer simply a medium through which democracy might expand; it is an economic and technological regime that actively shapes choices, attention, and action.

In this sense, Zuboff does not reject Castells' insights but historicises them. The network society created the conditions for global connectivity, yet early twenty-first-century computational tools reordered that space, transforming networks of communication into infrastructures of surveillance capitalism.

- 5) How would you analyse the impact of digitisation/automation of a public delivery system? Explain what parameters you would use to analyse and the reasons for choosing these parameters.